LEARNING OUTCOMES

- Identify input features and a labelled target for supervised learning.
- Inspect a dataset, carry out basic cleaning, and explain exploratory findings.
- Train a simple model without leaking test data into preprocessing.
- Evaluate predictions against a baseline and communicate limitations.
- Prepare a reproducible notebook and a detailed project report.

REQUIREMENTS:

1. Build a learning model based on the dataset assigned to each group.
2. Perform basic exploratory data analysis, basic data cleaning and feature preparation.
3. Train the model using 80-20 split with 42 as the random_state value, if applicable.
4. Use a confusion matrix for the performance evaluation of the model.
5. Each group must submit a complete documentation and the Python code that shows the ML processes.

DOCUMENTATION: The documentation must contain the following sections:
A. Introduction – It discusses what the problem is all about and the output of the project.

B. Data Collection and Preprocessing – It discusses the information about the dataset. Perform exploratory data analysis and interpret the results. Discuss the preprocessing techniques used and show the output after each technique, if applicable.

C. Feature Engineering – It discusses the feature engineering or feature selection methods used, if there is any. Provide justification for the selected features (you may use correlation analysis)

D. Data Splitting – It discusses the splitting value used for training and testing.

E. Model Selection and Training – It discusses the model or algorithm used for training and testing. Provide an explanation why you used that particular algorithm.

F. Model Evaluation – It discusses the performance of the model. Show the confusion matrix and interpret the result

GRADING CRITERIA:

Criterion	Max	Evidence required for full credit
Problem and dataset understanding	10	Clear objective, correct supervised task, source citation, target encoding, and data dictionary.
Basic cleaning	10	Before and after checks; justified actions; traceable cleaning log; no unnecessary deletion or invented issues.
Exploratory data analysis	15	At least three relevant, labelled plots; correct summaries; at least three evidence-based interpretations.
Feature preparation and splitting	10	Correct X and y, reproducible split, and training-only fitted preprocessing.
Model and baseline implementation	15	Appropriate simple algorithm and baseline; readable code; correct fitting and successful fresh-kernel execution.
Evaluation and interpretation	15	Correct required metrics, fair baseline comparison, labelled evaluation plots, error analysis, and sensible limitations.
Detailed documentation	10	Complete ML process, reasons for decisions, actual results, citations, and clear links to notebook evidence.
TOTAL	**85**	